

****These questions are basic questions to get you started with your studying.****

Chapter 3 Practice Problems

1. Ice is _____ dense than water because the hydrogen bonds are _____ in ice than in water.
 - a. Less dense, less ordered
 - b. More dense, more ordered
 - c. Less dense, more ordered
 - d. More dense, less ordered
2. Water molecules in ice are _____ than water molecules in liquid water
 - a. More spread out
 - b. Less spread out
3. Which of the following would you predict to dissolve best in water?
 - a. CO_2
 - b. H_2
 - c. NaCl
 - d. Xe
4. Which of the following is most hydrophilic?
 - a. CH_3OH
 - b. CH_4
 - c. CO_2
 - d. Fats & Oils
5. What does pH measure
 - a. H^+ concentration
 - b. OH^- concentration
 - c. The amount of energy in a solution
 - d. H_2O concentration
6. Which property of water makes it essential to life on earth?
 - a. The total amount of kinetic energy is has
 - b. It forms hydrogen bonds with other water molecules
 - c. It boils very quickly compared to other liquids
 - d. In its solid state, it absorbs carbon dioxide
7. Which of the following is an acid?
 - a. $[\text{H}^+] = 1 \times 10^{-14}$
 - b. $[\text{H}^+] = 1 \times 10^{-7}$
 - c. $[\text{H}^+] = 1 \times 10^{-9}$
 - d. $[\text{H}^+] = 1 \times 10^{-6}$
8. Which of the following represents the hydroxide ion concentration of pure water?

- a. $[H^+] = 1 \times 10^{-14}$
 - b. $[H^+] = 1 \times 10^{-7}$
 - c. $[H^+] = 1 \times 10^{-9}$
 - d. $[H^+] = 1 \times 10^{-6}$
9. If the pH of an aqueous solution is 5, what is the pOH?
10. In the above example, what is the $[H^+]$ concentration at pH 5?
11. Which solution has the greatest concentration of $[H^+]$?
- a. 1×10^{-4}
 - b. 1×10^{-10}
 - c. 1×10^{-1}
 - d. 1×10^{-7}
12. What type of bond explains the attraction of water molecules to other water molecules?
- a. Covalent bond
 - b. Ionic bond
 - c. Hydrogen bond
 - d. Iostopic bond
13. If solid water (ice) was more dense than liquid water, which of the following would occur?
- a. Humans, and other organisms that sweat, would not be able cool off properly
 - b. Most bodies of water would freeze in the winter and not completely thaw in the summer
 - c. Hurricanes would be much worse each summer
 - d. Hydrogen bonds would break more easily.
14. In water, the two hydrogen atoms are bonded to oxygen by what type of bonds?
- a. Ionic bond
 - b. Non-polar covalent bond
 - c. Polar covalent bond
 - d. Hydrogen bond

Answers:

- 1. C
- 2. A
- 3. C
- 4. A
- 5. A
- 6. B
- 7. D
- 8. B
- 9. 9

10. 1×10^{-5}

11. C

12. C

13. B

14. C